# Requirement: HR Analytics Report

*(Employee Attrition Analysis)*

## Project Objective

You are a **Power BI Developer** working for **NovaMed Innovations**. Your responsibility is to design and develop an interactive HR Analytics Report in Power BI that helps the HR department understand the key drivers of **employee attrition, analyze workforce trends and patterns, and support data-driven decision-making** to reduce attrition and improve employee retention across the organization.

## Expected Outcomes / Project Learnings

1. **Identify key factors** driving employee attrition.
2. **Improve the hiring process** by analyzing patterns in employee exits.
3. Enhance **employee experience** through data-backed insights.
4. Increase **workforce productivity** by addressing pain points.
5. Build **employee trust** through proactive retention strategies.

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## Scope of Work

### 1. Data Preparation (Power Query – Data Cleaning & Transformation)

- **Check and validate data types** for consistency across all fields.
- **Remove columns** containing any null values.
- **Eliminate duplicate records** to ensure data accuracy. (Expected rows 1470).
- The relevant columns for this report are:
  - ➔ **EmpID**
  - ➔ **Age**
  - ➔ **Age Group**
  - ➔ **Attrition**
  - ➔ **Department**
  - ➔ **Gender**
  - ➔ **Job Role**
  - ➔ **Job Satisfaction**
  - ➔ **Monthly Income**
  - ➔ **Salary Slab**
  - ➔ **YearAtCompany**

  (*You can hide the remaining columns.*)

### 2. DAX Calculations & Measures

- Create a measure to calculate **Total Employees, Total Attrition, Average Tenure, Average Salary, and Average Satisfaction Point**.
- Create a measure to calculate **Attrition Rate (%)** using the formula:
  *Attrition Rate (%) = (Total Attrition Count / Total Employee Count)*
  And convert into a percentage %.

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## 3. Report Visualizations

### Page 1: Attrition Overview

- **KPI Cards**:
  - Total Employees
  - Total Attrition
  - Attrition Rate (%)
  - Average Salary
  - Average Tenure
  - Avg Point Out of 4 (Average Satisfaction Point)
- **Visuals**:
  - **Gender-wise Attrition** → Donut or TreeMap chart showing attrition share between Male & Female.
  - **Attrition by Salary Slab** → Bar or Column chart comparing attrition across salary ranges.
  - **Attrition by Age Group** → Bar or Column chart for age-wise attrition distribution.
  - **Attrition by Tenure** → Line or Area chart showing employee tenure, with a filter pane to show the focus duration of years with the most attrition.
- **Slicers for Data Filtering**:
  - Department
  - Job Role

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## Page 2: Job Role & Satisfaction Analysis

- **KPI Cards**:
  - Total Employees
  - Total Attrition
  - Attrition Rate (%)
  - Average Salary
  - Average Tenure
  - Avg Point Out of 4 (Average Satisfaction Point)

- **Visuals**:
  - **Matrix Table:** Job Role vs. Job Satisfaction Score (restricted to employees who left).
  - **Attrition by Job Role**: Comparing attrition count by job role.
  - **Department-wise Attrition**: Bar or Column chart for department-wise attrition.

- **Slicers for Data Filtering**:
  - Department
  - Job Role

- **Navigation Button**: Add a button on **both pages** to toggle between **Page 1 (Overview)** and **Page 2 (Job Role & Satisfaction Analysis)**.

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